

1 UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

2 DOCTORAL THESIS

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4 **The study of Higgs properties in the**
5 **four-lepton final state at CMS**

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UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

Abstract

Institute of High Energy Physics
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Doctor of Science

The study of Higgs properties in the four-lepton final state at CMS

by Tahir JAVAID

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \rightarrow 4\ell$ decays ($\ell = e, \mu$) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV. The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to an integrated luminosity of 137.1 fb^{-1} at 13 TeV (Full Run 2 data) from the Large Hadron Collider (LHC) experiment.

The measurements are motivated from the theoretical predictions from well-known Standard Model (SM) of particle physics. After a long period of research, ordinary matter and underlying forces or interactions (electromagnetic, weak and strong interactions and not the gravity) which bind it in a structure are successfully explained by the SM. In SM, description of these interactions is made by symmetry laws of mathematics which could only explain the characteristic of particles with zero mass. Higgs mechanism was then introduced to make the theory realistic enough to incorporate mass problem. Under this mechanism, electroweak symmetry breaks and as a result particles gain the mass. Higgs boson is the direct manifestation of this Higgs mechanism [1, 2, 3, 4, 5, 6].

Precision measurements on the high energy physics experiments like Tevatron and LEP have proven the authenticity of SM; however, Higgs observation could not happen. In 2010, Large Hadron Collider (LHC), the largest experiment ever of its kind in physics world, was then set to put more tests on SM and it endorsed the measurements on SM from earlier experiments. LHC is located at CERN (European Organization for Nuclear Research) at Franco-Swiss border near Geneva, Switzerland and is most powerful setup of its kind to probe researches in particle physics. From its design, it is capable enough to accelerate and then collide two proton beams at center of mass energy upto 14 TeV at peak luminosity of $\mathcal{L} = 10^{34} \text{ cm}^2 \text{ s}^{-1}$. It is the most advanced, complex and largest experiment in the history of mankind, designed and constructed by a collaborative team of more than 10,000 technician, engineers and scientists from over 100 countries. It was constructed from 1998 to 2008, in a 3.8 meters wide tunnel of 27 Kilometers circumference at a depth which varies from 75-150 meters underground. This tunnel was previously used for LEP collider which was constructed from 1983 to 1988. Two counter rotating proton beams (anti-proton beam is disfavored to use) in two rings are used where the beam pipes are kept at ultra-high vacuum just to preserve the beams from possible extra collisions with molecules of gas. In 2012, the "hidden" Higgs boson, was discovered in shape of heavy scalar boson as jointly reported by the CMS and ATLAS experiments of LHC(CERN) [7, 8]. With this happening, physicists were all set to measure other properties of Higgs for further tests on SM predictions. Mass of Higgs boson was only free parameter in Higgs sector of SM until measured to be $m_H = 125.09 \pm 0.21(\text{stat.}) \pm 0.11(\text{syst.})$ GeV [9].

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) using the full Run 2 CMS data recorded in proton-proton

collisions at $\sqrt{s} = 13$ TeV are presented. Included Higgs differential measurements are based on Higgs kinematics and motivated from several aspects. For instance, Higgs transverse momentum can be used to test the pQCD calculations (especially when measured with exclusive multiplicities of jets). Additionally this variable is sensitive to Lagrangian structure of Higgs. Rapidity of Higgs is sensitive to parton distribution functions (PDFs) of the protons involved in the collisions. Jet related observables which are sensitive to theoretical modelling, boosted quark and also to the gluon emission are also covered in the dissertation. Current measurements are consistent with SM prediction; however, in such measurements, any deviation from SM expectations provides a direct hint to BSM contributions. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different decay channels of Higgs which can be studied in Ref. [Khachatryan:2015yvw, CMS:2015hja, CMS:2016rqf, Sirunyan:2017exp, CMS:2018hhg, ATLAS-CONF-2019-029, atlas-conf-2019-025, ATLAS:2018uso, ATLAS-CONF-2019-026, ATLAS:2017myr, CMS:2019chr, Sirunyan:2020xwk, 10, 11]. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

Integrated and differential fiducial cross section for $pp \rightarrow H \rightarrow 4\ell$ are measured by performing a maximum likelihood fit of the signal and background parameterisations to the observed 4ℓ mass distribution, $N_{\text{obs}}(m_{4\ell})$, and the fiducial cross section (σ_{fid}) is directly extracted from the fit. The systematic uncertainties are included in the form of nuisance parameters and are effectively integrated out in the fit procedure. The results are obtained using an asymptotic approach [12] with a test statistic based on the profile likelihood ratio [13]. This procedure for the unfolding of the detector effects from the observed distributions is the same as in Refs. [14] and [10]. In the case of the differential cross section measurements, the finite efficiencies and resolution effects are encoded in a detector response matrix which describes how events migrate from a given observable bin at the fiducial level to a given bin at the reconstruction level. This matrix is diagonally dominant, with sizeable off-diagonal elements for observables involving jets.

The dominant sources of systematic uncertainty are the experimental uncertainties in the lepton identification efficiencies and luminosity measurement, and the theoretical sources of uncertainty are found to be subdominant. In order to assess the model dependence of the measurement, the unfolding procedure is repeated using different response matrices created by varying the relative fraction of each SM production mode within its experimental constraints. The uncertainty is determined to be negligible with respect to the experimental systematic uncertainties.

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb at 13 TeV. The differential fiducial cross sections are presented as a function of several kinematic

115 observables which are sensitive to the Higgs-boson production mechanisms: ra-
 116 pidity and transverse momentum of the four-lepton system, associated jet multi-
 117 plicity and transverse momentum of the leading jet. The differential measurements
 118 are done in bins of the observables. For each kinematic or differential observable,
 119 selection procedure of the binning boundaries has been optimized in order to have
 120 a similar expected measured cross-section for every bin of the observable. For the
 121 differential cross section measurements, the background is to be subtracted at the
 122 reconstruction level for each bin of the observable. The mass spectrum for the
 123 background is not same for all differential bins because it does not have a reso-
 124 nance structure. For the irreducible and reducible backgrounds, the four-lepton
 125 mass spectra are obtained as the $m_{4\ell}$ template shapes through simulated events
 126 and the control regions in the data respectively for each bin of the observable for
 127 differential measurements as the procedure is described in Ref. [15]. These mea-
 128 surements are important to test the Standard Model predictions in the Higgs sector
 129 and can be used to constrain phase spaces of many theories beyond the Standard
 130 Model.

131 All the integrated and differential measurements are compared with the theo-
 132 retical predictions based on the Standard Model and found to be consistent within
 133 their uncertainties. More precise Higgs boson properties measurements are needed
 134 to spot any possible deviation from SM predictions which can be a doorway to new
 135 physics.

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